Part VII

Choice Under Uncertainty 1

Why study uncertainty?

So far we have covered individual decision-making under certainty

- Goods well understood
- Prices well known

In fact, decisions typically made in the face of an uncertain future

- Workhorse model: objective risk
- Subjective assessments of uncertainty
- Behavioral critiques

von Neumann-Morganstern expected utility model

Simplifying assumptions include

- Finite number of outcomes ("prizes")
- Objectively known probability distributions over prizes ("lotteries")
- Complete and transitive preferences over lotteries
  - Other assumptions on preferences over lotteries (to be discussed)

Outline

- Uncertainty setup
  - Prizes and lotteries
  - Preferences
- Expected utility representation
- Lotteries with monetary payoffs
- Measuring risk aversion
  - Certain equivalent
  - Arrow-Pratt coefficient of absolute risk aversion
  - Risk preferences and wealth

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Prizes and lotteries

Let $\mathcal{X}$ be the set of possible prizes (a.k.a. outcomes or consequences)

- Assume $|\mathcal{X}| = n < \infty$
- Since $|\mathcal{X}| < \infty$, there must be a best outcome and a worst outcome

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A lottery is a probability distribution over prizes

- $p = (p_1, \ldots, p_n) \in \mathbb{R}^n_+$
- The set of all lotteries is

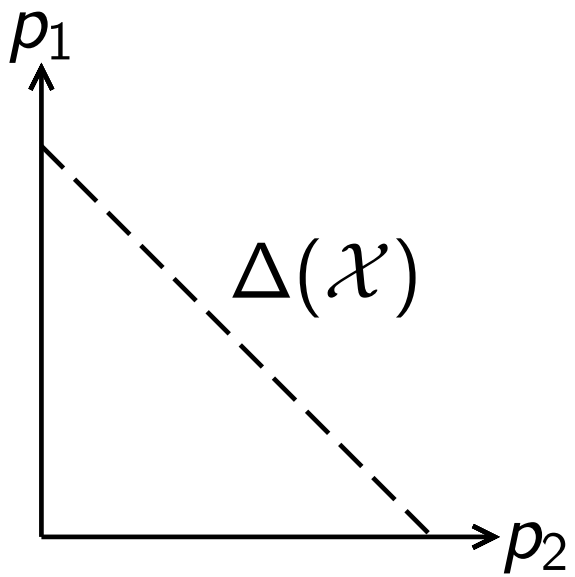
$$\Delta(\mathcal{X}) \equiv \left\{ p \in \mathbb{R}^n_+ : \sum_i p_i = 1 \right\},$$

called the n-dimensional simplex

Graphing the simplex $\Delta(\mathcal{X}) \subseteq \mathbb{R}^2$

Suppose there are two prizes ($|\mathcal{X}| = 2$)

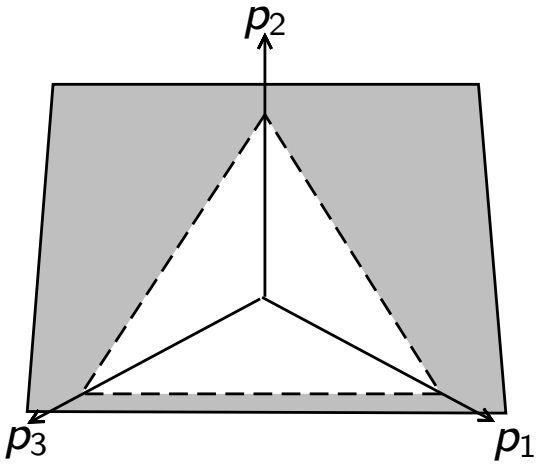
- The simplex $\Delta(\mathcal{X})$ is the portion of the line $p_1 + p_2 = 1$ that lies in the positive quadrant
- This is a one-dimensional submanifold of two-dimensional space—we can draw it as a line segment (i.e., an interval)



Graphing the simplex $\Delta(\mathcal{X}) \subseteq \mathbb{R}^3$

Suppose there are three prizes ($|\mathcal{X}| = 3$)

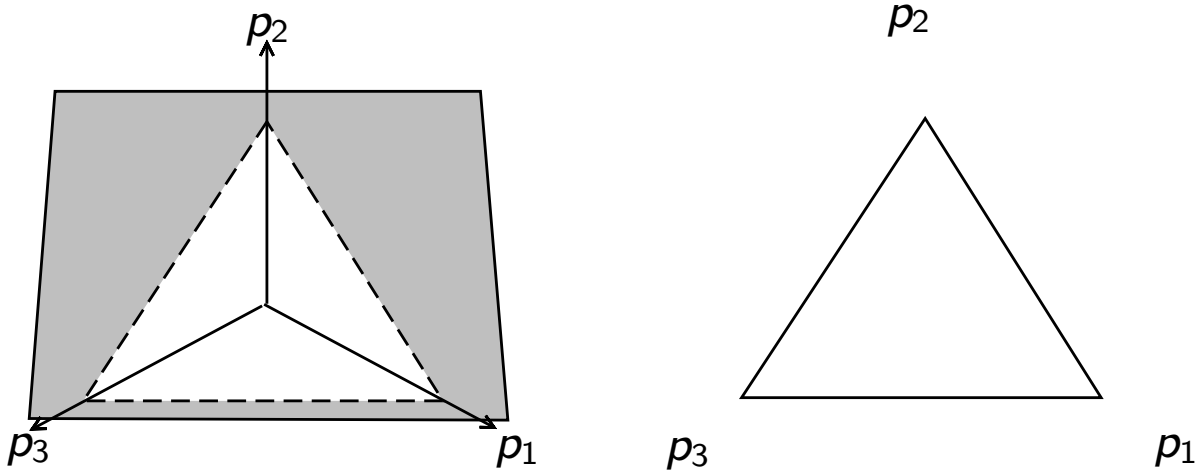
- The simplex $\Delta(\mathcal{X})$ is the portion of the plane $p_1 + p_2 + p_3 = 1$ that lies in the positive orthant
- This is a two-dimensional submanifold of three-dimensional space



Graphing the simplex $\Delta(\mathcal{X}) \subseteq \mathbb{R}^3$

Suppose there are three prizes ($|\mathcal{X}| = 3$)

- The simplex $\Delta(\mathcal{X})$ is the portion of the plane $p_1 + p_2 + p_3 = 1$ that lies in the positive orthant
- This is a two-dimensional submanifold of three-dimensional space—we can draw it as a triangle



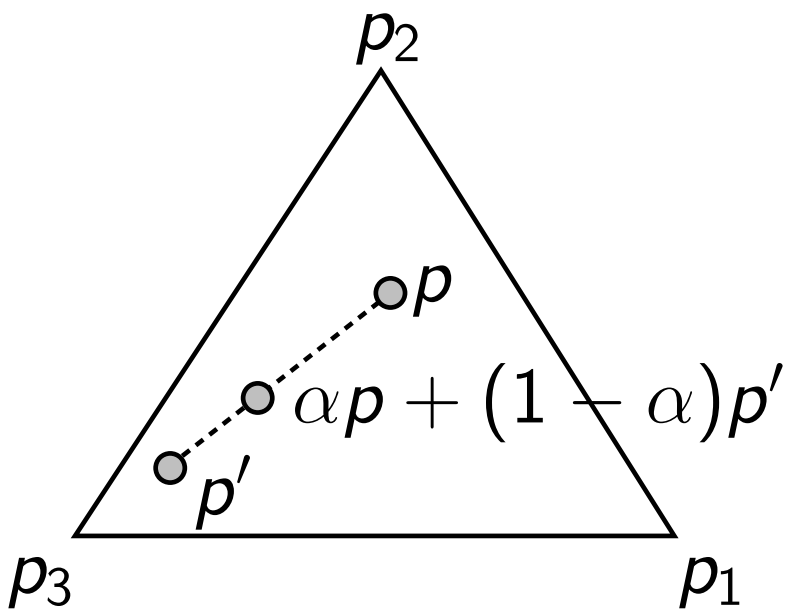
- Aside: These drawings are said to use "barycentric coordinates"

Convexity of the simplex I

Note that $\Delta(\mathcal{X})$ is a convex set

- If $p_i \geq 0$ and $p'_i \geq 0$, then $\alpha p_i + (1-\alpha)p'_i \geq 0$
- If $\sum_i p_i = 1$ and $\sum_i p'_i = 1$, then $\sum_i [\alpha p_i + (1-\alpha)p'_i] = 1$

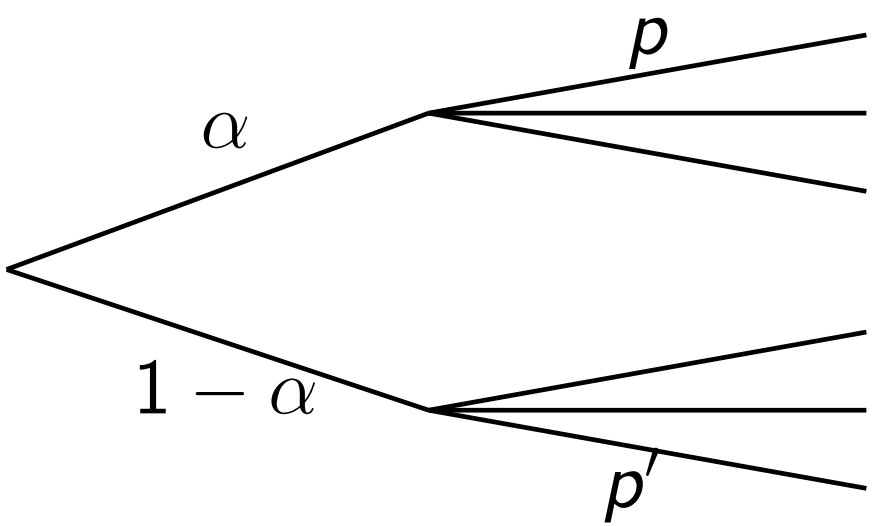
This is not surprising given that the simplex is a "triangle"



Convexity of the simplex II

We can view $\alpha p + (1-\alpha)p'$ as a compound lottery

1. Choose between lotteries: Lottery p with probability α and lottery p' with probability $(1-\alpha)$
2. Resolve uncertainty in chosen lottery per p or p'



Preferences over lotteries

A rational decision-maker has preferences over outcomes $\mathcal{X}$

We consider preferences over lotteries $\Delta(\mathcal{X})$ (note that from here on, $\succsim$ refers to preferences over lotteries, not outcomes)

Expected utility theory relies on $\succsim$ satisfying

- Completeness
- Transitivity
- Continuity (in a sense to be defined)
- Independence (to be defined)

Continuity axiom

Definition (continuity)

A preference relation $\succsim$ over $\Delta(\mathcal{X})$ is continuous iff for any p_H, p_M, and $p_L \in \Delta(\mathcal{X})$ such that $p_H \succsim p_M \succsim p_L$, there exists some $\alpha \in [0, 1]$ such that

$$\alpha p_H + (1 - \alpha) p_L \sim p_M.$$

Independence axiom

Definition (independence)

A preference relation $\succsim$ over $\Delta(\mathcal{X})$ satisfies independence iff for any p, p', and $p_m \in \Delta(\mathcal{X})$ and any $\alpha \in [0,1]$, we have

$$p \succsim p'$$
$$\Updownarrow$$
$$\alpha p + (1-\alpha)p_m \succsim \alpha p' + (1-\alpha)p_m.$$

i.e., if I prefer p to p', I also prefer the possibility of p to the possibility of p', as long as the other possibility is the same (a $(1-\alpha)$ chance of p_m) in both cases

Independence sensible for choice under uncertainty

There is no counterpart in standard consumer theory; e.g.,

- $p = (2 \text{ coke}, 0 \text{ twinkies})$ and $p' = (0 \text{ coke}, 2 \text{ twinkies})$
- $p_m = (2 \text{ coke}, 2 \text{ twinkies})$
- $\alpha = \frac{1}{2}$

$$\overbrace{(2 \text{ coke}, 0 \text{ twinkies})}^{p} \succsim \overbrace{(0 \text{ coke}, 2 \text{ twinkies})}^{p'}$$

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- $\alpha = \frac{1}{2}$

There is no reason to conclude that

$$\overbrace{(2 \text{ coke}, 0 \text{ twinkies})}^{p} \succsim \overbrace{(0 \text{ coke}, 2 \text{ twinkies})}^{p'}$$

$$\Updownarrow$$

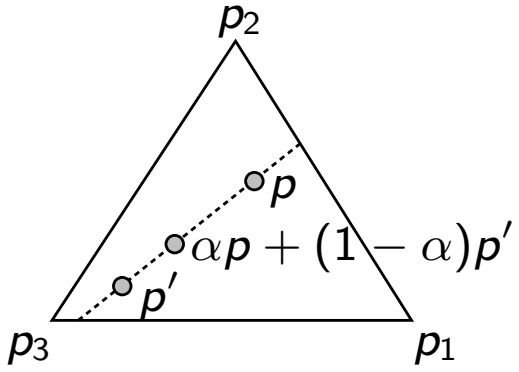
$$\underbrace{(2 \text{ coke}, 1 \text{ twinkies})}_{\alpha p + (1-\alpha) p_m} \succsim \underbrace{(1 \text{ coke}, 2 \text{ twinkies})}_{\alpha p' + (1-\alpha) p_m}$$

Independence implies linear indifference curves

Independence implies linear indifference curves

- Consider $p \sim p'$
- Let $p_m = p'$
- By the independence axiom,

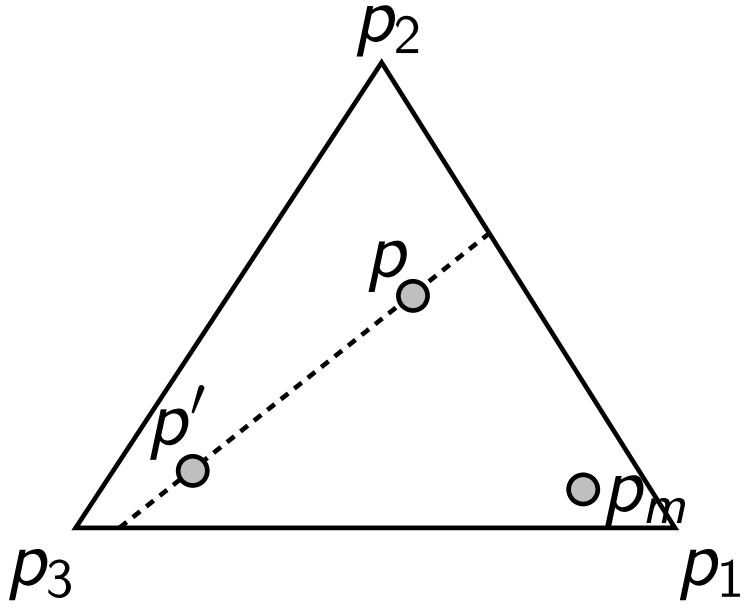
$$\begin{aligned} \alpha p + (1-\alpha)p' &\sim \alpha p' + (1-\alpha)p' \\ &\sim p' \\ &\sim p \end{aligned}$$



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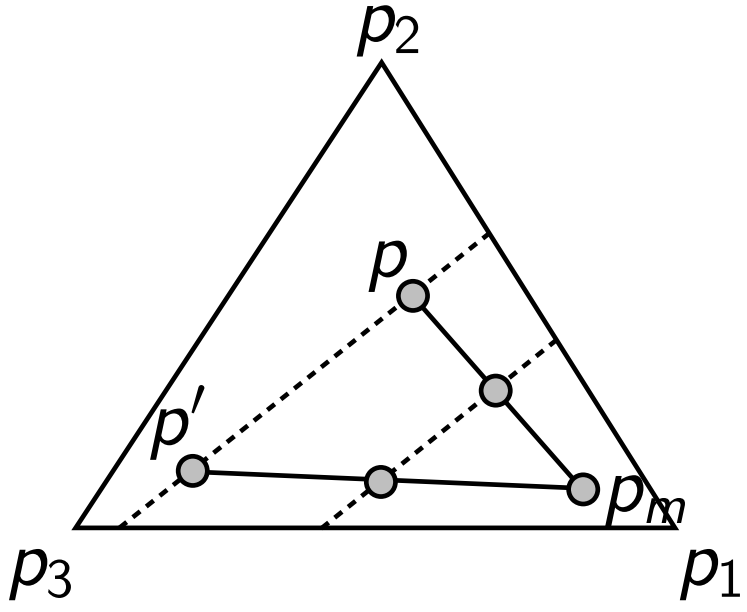
- Consider $p \sim p'$
- Let p_m be some other point, and α some value in $(0, 1)$
- By independence $\alpha p + (1 - \alpha)p_m \sim \alpha p' + (1 - \alpha)p_m$
- $\alpha p + (1 - \alpha)p_m$ and $\alpha p' + (1 - \alpha)p_m$ lie on a line parallel to the indifference curve containing p and p'



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von Neumann-Morganstern utility functions

Definition (von Neumann-Morganstern utility function)

A utility function $U\colon \Delta(\mathcal{X}) \to \mathbb{R}$ is a vNM utility function iff there exist numbers $u_1, \ldots, u_n \in \mathbb{R}$ such that for every $p \in \Delta(\mathcal{X})$,

$$U(p) = \sum_{i=1}^{n} p_i u_i = p \cdot \vec{u}.$$

Can think of $u_1, \ldots, u_n$ as indexing preference over outcomes

Linearity of vNM utility functions I

Theorem

A utility function $U\colon \Delta(\mathcal{X}) \to \mathbb{R}$ *is a vNM utility function iff it is linear in probabilities, i.e.,*

$$U\big(\alpha p + (1-\alpha)p'\big) = \alpha U(p) + (1-\alpha)U(p')$$

for all p, $p' \in \Delta(\mathcal{X})$, *and* $\alpha \in [0,1]$

Linearity of vNM utility functions II

Proof.

vNM $\implies$ linearity:

$$\begin{aligned} U\big(\alpha p + (1-\alpha)p'\big) &= \big(\alpha p + (1-\alpha)p'\big) \cdot \vec{u} \\ &= (\alpha p) \cdot \vec{u} + \big((1-\alpha)p'\big) \cdot \vec{u} \\ &= \alpha(p \cdot \vec{u}) + (1-\alpha)(p' \cdot \vec{u}) \\ &= \alpha U(p) + (1-\alpha)U(p') \end{aligned}$$

vNM $\impliedby$ linearity: not shown here $\square$

Expected utility representation and ordinality

If preferences $\succsim$ can be represented by a vNM utility function, we say it is an "expected utility representation" of $\succsim$
That is "$U(\cdot)$ is an expected utility representation of $\succsim$" means

1. $U(\cdot)$ is a vNM utility function, and
2. $U(\cdot)$ represents $\succsim$

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Linearity of vNM utility functions mean that expected utility representation is not ordinal

- Utility representation is robust to any increasing monotone transformation
- Expected utility representation is only robust to affine (increasing linear) transformations

Exp. util. representation robust to affine transformation I

Theorem

Suppose $U\colon \Delta(\mathcal{X}) \to \mathbb{R}$ *is an expected utility representation of* $\succsim$. *Then* $V\colon \Delta(\mathcal{X}) \to \mathbb{R}$ *is also an expected utility representation of* $\succsim$ *iff there exist some* $a \in \mathbb{R}$ *and* $b \in \mathbb{R}_{++}$ *such that*

$$V(p) = a + bU(p)$$

for all $p \in \Delta(\mathcal{X})$.

Exp. util. representation robust to affine transformation II

Proof.

Suppose $U(p) = p \cdot \vec{u}$.

$V(p) = a + bU(p) \Longrightarrow V$ expected utility represents $\succsim$:

- V represents $\succsim$ since $b > 0$ gives that

$$U(p') \geq U(p) \implies a + bU(p') \geq a + bU(p)$$
$$V(p') \geq V(p).$$

- V is vNM utility function since

$$V(p) = a + bU(p) = a + b\sum_{i=1}^{n} p_i u_i = \sum_{i=1}^{n} p_i \underbrace{(a + bu_i)}_{\equiv v_i} = p \cdot \vec{v}.$$

Exp. util. representation robust to affine transformation III

Proof (continued).

$V(p) = a + bU(p) \Longleftarrow V$ expected utility represents $\succsim$:

- Let $\underline{p}$ be the worst lottery (the one giving the worst outcome for certain), and let $\bar{p}$ be the best lottery (the one giving the best outcome for certain). Suppose $\bar{p} \succ \underline{p}$ (if $\bar{p} \sim \underline{p}$ the result is trivial).
- For every $p \in \Delta(\mathcal{X})$, we have $\bar{p} \succsim p \succsim \underline{p}$. Thus $U(\bar{p}) \geq U(p) \geq U(\underline{p})$ so there exists some $\lambda_p \in [0, 1]$ such that

$$U(p) = \lambda_p U(\bar{p}) + (1 - \lambda_p) U(\underline{p}),$$

in particular,

$$\lambda_p = \frac{U(p) - U(\underline{p})}{U(\bar{p}) - U(\underline{p})}.$$

Exp. util. representation robust to affine transformation IV

Proof (continued).

- We have

$$\begin{aligned} U(p) &= \lambda_p U(\bar{p}) + (1-\lambda_p) U(\underline{p}) \\ &= U\big(\lambda_p \bar{p} + (1-\lambda_p)\underline{p}\big) \\ p &\sim \lambda_p \bar{p} + (1-\lambda_p)\underline{p}. \end{aligned}$$

Since V expected utility represents $\succsim$,

$$\begin{aligned} V(p) &= V\big(\lambda_p \bar{p} + (1-\lambda_p)\underline{p}\big) \\ &= \lambda_p V(\bar{p}) + (1-\lambda_p) V(\underline{p}). \end{aligned}$$

Exp. util. representation robust to affine transformation V

Proof (continued).

- Define

$$a \equiv V(\underline{p}) - U(\underline{p})b \quad \text{and} \quad b \equiv \frac{V(\bar{p}) - V(\underline{p})}{U(\bar{p}) - U(\underline{p})}.$$

- We seek to show that $V(p) = a + bU(p)$

$$\begin{aligned} a + bU(p) &= V(\underline{p}) - U(\underline{p})b + bU(p) \\ &= V(\underline{p}) + b\big[U(p) - U(\underline{p})\big] \\ &= V(\underline{p}) + \underbrace{\frac{U(p) - U(\underline{p})}{U(\bar{p}) - U(\underline{p})}}_{\lambda_p}\big[V(\bar{p}) - V(\underline{p})\big] \\ &= \lambda_p V(\bar{p}) + (1 - \lambda_p)V(\underline{p}) \\ &= V(p). \end{aligned}$$

□

Level sets of an expected utility function

A von Neumann-Morganstern utility function satisfies

$$U(p) = \sum_{i=1}^{n} p_i u_i = p \cdot \vec{u}$$

for some $\vec{u} \in \mathbb{R}^n$

Indifference curves are therefore $p \cdot \vec{u} = c$ for various c

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- Straight lines ($p \cdot \vec{u} = c$ is a plane that intercepts the simplex in a line)

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Indifference curves are therefore

- Straight lines ($p \cdot \vec{u} = c$ is a plane that intercepts the simplex in a line)
- Parallel (all indifference curves are normal to $\vec{u}$)

Which preferences have expected utility representations? I

Theorem

A complete and transitive preference relation $\succsim$ on $\Delta(\mathcal{X})$ satisfies continuity and independence iff it has an expected utility representation $U\colon \Delta(\mathcal{X}) \to \mathbb{R}$.

Showing that if $U(p) = p \cdot \vec{u}$ represents $\succsim$, then $\succsim$ must satisfy continuity and independence is (relatively) easy

Showing the other direction is a bit harder...

Which preferences have expected utility representations? II

Formal proof of other direction is given in notes; roughly:

1. For every $p \in \Delta(\mathcal{X})$ find $\lambda_p \in [0,1]$ such that

$$p \sim \lambda_p \bar{p} + (1 - \lambda_p)\underline{p}$$

   This λ_p...
   - exists by continuity
   - is unique by independence

2. Let $U(p) = \lambda_p$
3. Show that $U(\cdot)$ is an expected utility representation of $\succsim$
   - $U(\cdot)$ represents $\succsim$
   - $U(\cdot)$ is linear, and therefore is a vNM utility function

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Money lotteries

We seek to measure the attitude towards risk embedded in preferences $\succsim$

To make things easier, we limit outcomes to monetary payoffs

- $\mathcal{X} \subseteq \mathbb{R}$ (note we give up assumption that prize set is finite)
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Money lotteries

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To make things easier, we limit outcomes to monetary payoffs

- $\mathcal{X} \subseteq \mathbb{R}$ (note we give up assumption that prize set is finite)
- $\Delta(\mathcal{X})$ is now a bit more complicated
  - A probability distribution with finite support can be described with a pmf; set of distributions is the simplex
  - A probability distribution over infinite (ordered) support is described by a cdf

From here on, we'll represent a lottery by a cdf $F(\cdot)$, where $F(x)$ is the probability of receiving less than or equal to an $\$x$ payout

What is the set of lotteries?

When $|\mathcal{X}| = n < \infty$, the set of all lotteries is

$$\Delta(\mathcal{X}) \equiv \left\{ p \in \mathbb{R}^n_+ : \sum_i p_i = 1 \right\}$$

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When $|\mathcal{X}| = n < \infty$, the set of all lotteries is

$$\Delta(\mathcal{X}) \equiv \left\{ p \in \mathbb{R}^n_+ : \sum_i p_i = 1 \right\}$$

When $\mathcal{X} = \mathbb{R}$, the set of all lotteries is the set of cdfs:
$\mathbb{F}$ is the set of all functions $F : \mathbb{R} \to [0, 1]$ such that

- $F(\cdot)$ is nondecreasing
- $\lim_{x \to -\infty} F(x) = 0$
- $\lim_{x \to +\infty} F(x) = 1$
- $F(\cdot)$ is right-continuous (i.e., $\lim_{h \downarrow 0} F(x + h) = F(x)$ for all x)

Preferences over money lotteries

Our old vNM utility function (over pmfs) was

$$U(p) = \sum_i p_i u_i \equiv \mathbb{E}_p u$$

Preferences over money lotteries

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The continuous analogue is a vNM utility function over cdfs:

$$U(F) = \int_{\mathbb{R}} u(x)\, dF(x) \equiv \mathbb{E}_F\big[u(x)\big]$$

Where

- $U\colon \mathbb{F} \to \mathbb{R}$ ("von Neumann-Morganstern utility function") represents preferences over lotteries
- $u\colon \mathbb{R} \to \mathbb{R}$ ("Bernoulli utility function") indexes preference over outcomes

Risk aversion I

Definition (risk aversion)

A decision-maker is risk-averse iff for all lotteries F, she prefers a certain payoff of $\mathbb{E}_F(x) \equiv \int_{\mathbb{R}} x\, dF(x)$ to the lottery F.

Definition (strict risk aversion)

A decision-maker is strictly risk-averse iff for all non-degenerate lotteries F (i.e, all lotteries for which the support of F is not a singleton), she strictly prefers a certain payoff of $\mathbb{E}_F(x) \equiv \int_{\mathbb{R}} x\, dF(x)$ to the lottery F.

Risk aversion II

Risk aversion says that for all F,

$$u\big(\mathbb{E}_F[x]\big) \geq \mathbb{E}_F\big[u(x)\big]$$

or equivalently

$$u\left(\int_{\mathbb{R}} x\, dF(x)\right) \geq \int_{\mathbb{R}} u(x)\, dF(x)$$

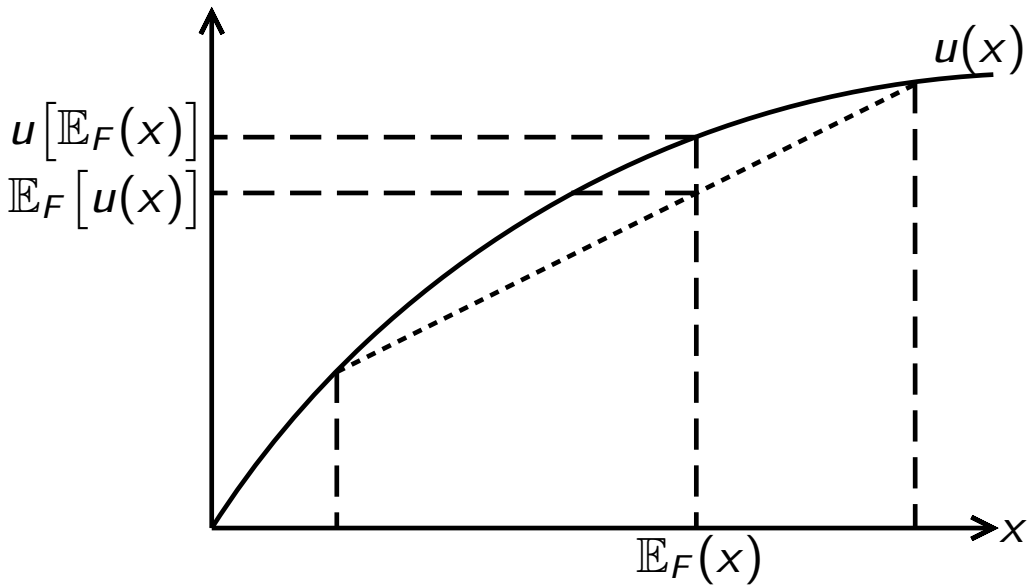
By Jensen's inequality, this condition holds iff $u(\cdot)$ is concave

Theorem

A decision-maker is (strictly) risk-averse iff her Bernoulli utility function is (strictly) concave.

Illustrating risk aversion

Consider a risk-averse decision-maker (i.e., one with a concave Bernoulli utility function) evaluating a lottery F with a two-point distribution



$$u\big(\mathbb{E}_F[x]\big) \geq \mathbb{E}_F\big[u(x)\big]$$

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The certain equivalent

A risk-averse decision-maker prefers a certain payoff of $\mathbb{E}_F(x)$ to the lottery F

$$u\big(\mathbb{E}_F[x]\big) \geq \mathbb{E}_F\big[u(x)\big]$$

How many "certain" dollars is F worth? That is, what is the certain payoff that gives the same utility as lottery F?

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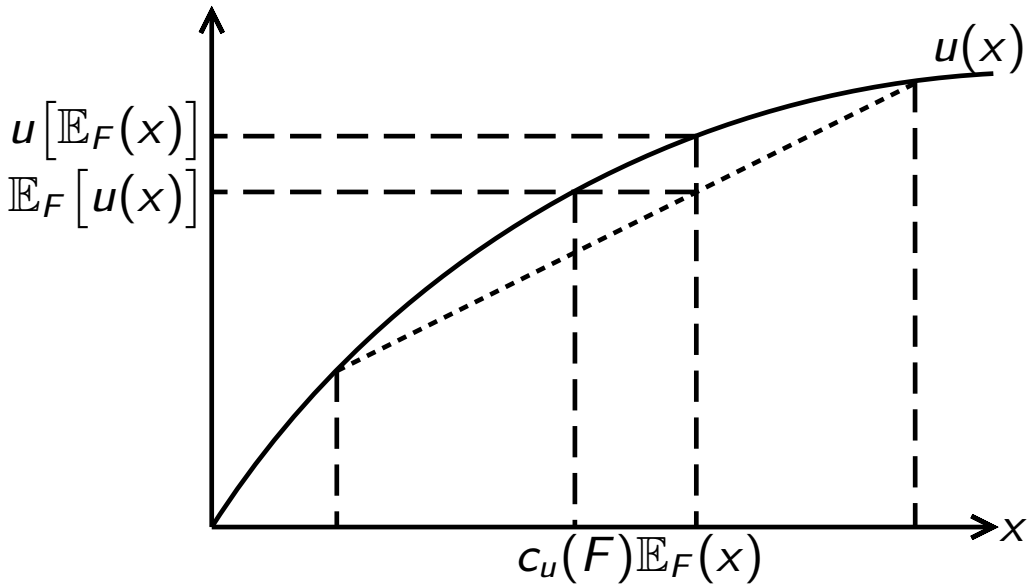
Definition (certain equivalent)

The certain equivalent is the size of the certain payout such that a decision-maker is indifferent between the certain payout and the lottery F:

$$u\big(c_u(F)\big) = \mathbb{E}_F\big[u(x)\big] \equiv \int_{\mathbb{R}} u(x)\, dF(x).$$

Illustrating the certain equivalent

Consider a risk-averse decision-maker (i.e., one with a concave Bernoulli utility function) evaluating a lottery F with a two-point distribution



$$u\big(c_u(F)\big) = \mathbb{E}_F\big[u(x)\big] \leq u\big(\mathbb{E}_F[x]\big)$$

The certain equivalent as a measure of risk aversion

For a risk-averse decision-maker

$$u\big(c_u(F)\big) = \mathbb{E}_F\big[u(x)\big] \leq u\big(\mathbb{E}_F[x]\big)$$

so assuming increasing $u(\cdot)$,

$$c_u(F) \leq \mathbb{E}_F[x]$$

The certain equivalent gives a measure of risk aversion

- A consumer u is risk-averse iff $c_u(F) \leq \mathbb{E}_F[x]$ for all F
- Consumer u is *more* risk-averse than consumer v iff $c_u(F) \leq c_v(F)$ for all F

The Arrow-Pratt coefficient of absolute risk aversion

Definition (Arrow-Pratt coefficient of absolute risk aversion)

Given a twice differentiable Bernoulli utility function $u(\cdot)$,

$$A_u(x) \equiv -\frac{u''(x)}{u'(x)}.$$

Where does this come from?

- Risk-aversion is related to concavity of $u(\cdot)$; a "more concave" function has a smaller second derivative hence a larger $-u''(x)$
  - Normalization by $u'(x)$ takes care of the fact that $au(\cdot) + b$ represents the same preferences as $u(\cdot)$
- We can also view it as a "probability premium"

The A-P coefficient of ARA as a probability premium

Consider a risk-averse consumer:

- She prefers x for certain to a 50-50 gamble between $x + \varepsilon$ and $x - \varepsilon$

The A-P coefficient of ARA as a probability premium

Consider a risk-averse consumer:

- She prefers x for certain to a 50-50 gamble between $x + \varepsilon$ and $x - \varepsilon$
- If we wanted to convince her to take such a gamble, it couldn't be 50-50—we need to make the $x + \varepsilon$ payout more likely

The A-P coefficient of ARA as a probability premium

Consider a risk-averse consumer:

- She prefers x for certain to a 50-50 gamble between $x + \varepsilon$ and $x - \varepsilon$
- If we wanted to convince her to take such a gamble, it couldn't be 50-50—we need to make the $x + \varepsilon$ payout more likely
- Consider the gamble G such that she is indifferent between G and receiving x for certain, where

$$G = \begin{cases} x + \varepsilon & \text{with probability } \frac{1}{2} + \pi, \\ x - \varepsilon & \text{with probability } \frac{1}{2} - \pi \end{cases}$$

The A-P coefficient of ARA as a probability premium

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- It turns out that $A_u(x)$ is proportional to (π/ε) as $\varepsilon \to 0$; i.e., $A_u(x)$ tells us the "premium" measured in probability that the decision-maker demands per unit of spread ε

Our measures of risk aversion are equivalent I

Theorem

The following definitions of u being "more risk-averse" than v are equivalent:

1. *Whenever u prefers F to a certain payout d, then v does as well; i.e., for all F and d,*

$$\mathbb{E}_F\big[u(x)\big] \geq u(d) \implies \mathbb{E}_F\big[v(x)\big] \geq v(d);$$

2. *Certain equivalents $c_u(F) \leq c_v(F)$ for all F;*
3. *$u(\cdot)$ is "more concave" than $v(\cdot)$; i.e., there exists some increasing concave function $g(\cdot)$ such that $u(x) = g\big(v(x)\big)$ for all x;*
4. *Arrow-Pratt coefficients of absolute risk aversion $A_u(x) \geq A_v(x)$ for all x.*

Our measures of risk aversion are equivalent II

Proof.

$1 \Leftrightarrow 2$: Suppose that 1 does not hold; i.e., there exists some F and d such that

$$\begin{aligned} \mathbb{E}_F\big[u(x)\big] &\geq u(d) & \qquad \mathbb{E}_F\big[v(x)\big] &< v(d) \\ u\big[c_u(F)\big] &\geq u(d) & \qquad v\big[c_v(F)\big] &< v(d) \\ c_u(F) &\geq d & \qquad c_v(F) &< d. \end{aligned}$$

Thus $c_u(F) > c_v(F)$. Implication also goes from bottom to top (if there is some F for which $c_u(F) > c_v(F)$, then there must be some d such that $c_u(F) \geq d > c_v(F)$).

Our measures of risk aversion are equivalent III

Proof (continued).

2 $\Leftrightarrow$ 3: u and v are both monotone functions, so there is some increasing function g such that $u(x) = g\big(v(x)\big)$ for all x.

$$
\begin{aligned}
c_u(F) &\leq c_v(F) \\
&\Updownarrow \\
u\big[c_u(F)\big] &\leq u\big[c_v(F)\big] \\
\mathbb{E}_F\big[u(x)\big] &\leq \\
\mathbb{E}_F\big[g\big(v(x)\big)\big] &\leq g\big(v\big[c_v(F)\big]\big) \\
&\leq g\big(\mathbb{E}_F\big[v(x)\big]\big).
\end{aligned}
$$

This holds for all F iff $g(\cdot)$ is concave by Jensen's inequality.

Our measures of risk aversion are equivalent IV

Proof (continued).

3 $\Leftrightarrow$ 4: Existence of $A_u(x)$ and $A_v(x)$ presupposes differentiability.

$$
\begin{aligned}
u(x) &= g\big(v(x)\big) \\
u'(x) &= g'\big(v(x)\big)v'(x) \\
u''(x) &= g'\big(v(x)\big)v''(x) + g''\big(v(x)\big)\big(v'(x)\big)^2
\end{aligned}
$$

$$
\begin{aligned}
A_u(x) = -\frac{u''(x)}{u(x)} &= -\frac{\cancel{g'\big(v(x)\big)}v''(x)}{\cancel{g'\big(v(x)\big)}v'(x)} - \frac{g''\big(v(x)\big)\big(v'(x)\big)^{\cancel{2}}}{g'\big(v(x)\big)\cancel{v'(x)}} \\
&= A_v(x) - g''\big(v(x)\big)\frac{v'(x)}{g'\big(v(x)\big)}.
\end{aligned}
$$

Since $v(\cdot)$ and $g(\cdot)$ are increasing functions, we have $A_u(x) \geq A_v(x)$ for all x iff $g'' \leq 0$ for all x. $\square$

Decreasing absolute risk aversion

Definition (decreasing absolute risk aversion)

The Bernoulli utility function $u(\cdot)$ has decreasing absolute risk aversion iff $A_u(\cdot)$ is a decreasing function of x.

Decreasing absolute risk aversion

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Definition (increasing absolute risk aversion)

The Bernoulli utility function $u(\cdot)$ has increasing absolute risk aversion iff $A_u(\cdot)$ is an increasing function of x.

Definition (constant absolute risk aversion)

The Bernoulli utility function $u(\cdot)$ has constant absolute risk aversion iff $A_u(\cdot)$ is a constant function of x.

Relative risk aversion

Definition (coefficient of relative risk aversion)

Given a twice differentiable Bernoulli utility function $u(\cdot)$,

$$R_u(x) \equiv -x\frac{u''(x)}{u'(x)} = xA_u(x).$$

Relative risk aversion

Definition (coefficient of relative risk aversion)

Given a twice differentiable Bernoulli utility function $u(\cdot)$,

$$R_u(x) \equiv -x\frac{u''(x)}{u'(x)} = xA_u(x).$$

We can define decreasing/increasing/constant relative risk aversion as above, but using $R_u(\cdot)$ instead of $A_u(\cdot)$

- DARA means that if I take a \$10 gamble when poor, I will take a \$10 gamble when rich
- DRRA means that if I gamble 10% of my wealth when poor, I will gamble 10% when rich

Part VIII

Choice Under Uncertainty 2

Outline

- Comparing risky prospects
  - First-order stochastic dominance
  - Second-order stochastic dominance
- Application: demand for insurance
- Application: the portfolio problem
- Beyond the expected utility model
  - Subjective probability
  - Behavioral criticisms

Outline

- Comparing risky prospects
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When is one lottery "better than" another?

We can compare lotteries given a Bernoulli utility function: $\succsim_u$

When will two lotteries be consistently ranked under a broad set of preferences? e.g.,

1. All nondecreasing $u(\cdot)$
2. All nondecreasing, concave (i.e., risk averse) $u(\cdot)$

Comparing lotteries: examples I

Example

\$95 for certain vs. \$105 for certain.

Example

$$\begin{pmatrix} \$90 & \text{with probability } 1/2 \\ \$110 & \text{with probability } 1/2 \end{pmatrix} \text{ vs. } \$95 \text{ for certain.}$$

Example

$$\begin{pmatrix} \$90 & \text{with probability } 1/2 \\ \$110 & \text{with probability } 1/2 \end{pmatrix} \text{ vs. } \$105 \text{ for certain.}$$

Comparing lotteries: examples II

Example

$$\begin{pmatrix} \$90 & \text{with probability } 1/2 \\ \$110 & \text{with probability } 1/2 \end{pmatrix} \text{ vs. } \$110 \text{ for certain.}$$

Example

$$\begin{pmatrix} \$90 & \text{with probability } 1/2 \\ \$110 & \text{with probability } 1/2 \end{pmatrix} \text{ vs. } \begin{pmatrix} \$80 & \text{with probability } 1/2 \\ \$120 & \text{with probability } 1/2 \end{pmatrix}.$$

First-order stochastic dominance

Definition (first-order stochastic dominance)

Distribution G first-order stochastic dominates distribution F iff lottery G is preferred to F under every nondecreasing Bernoulli utility function $u(\cdot)$.

That is, for every nondecreasing $u\colon \mathbb{R} \to \mathbb{R}$, the following (equivalent) statements hold:

$$G \succsim_u F,$$

$$\mathbb{E}_G\big[u(x)\big] \geq \mathbb{E}_F\big[u(x)\big],$$

$$\int_{\mathbb{R}} u(x)\, dG(x) \geq \int_{\mathbb{R}} u(x)\, dF(x).$$

Characterizing first-order stochastic dominant cdfs I

Theorem

Distribution G first-order stochastic dominates distribution F iff $G(x) \leq F(x)$ for all x.

That is, lottery G is more likely than F to pay at least x for any threshold x

Characterizing first-order stochastic dominant cdfs II

Proof.

We assume differentiability so we can integrate by parts:

$$\int_{\mathbb{R}} u(x)\,dG(x) = u(x)G(x)\Big|_{x=-\infty}^{x=\infty} - \int_{\mathbb{R}} u'(x)G(x)\,dx$$
$$\int_{\mathbb{R}} u(x)\,dF(x) = u(x)F(x)\Big|_{x=-\infty}^{x=\infty} - \int_{\mathbb{R}} u'(x)F(x)\,dx.$$

Note that $\lim_{x\to-\infty} u(x)G(x) = \lim_{x\to-\infty} u(x)F(x) = 0$. Assume that $\lim_{x\to+\infty} u(x)[G(x) - F(x)] = 0$, so

$$\int_{\mathbb{R}} u(x)\,dG(x) - \int_{\mathbb{R}} u(x)\,dF(x) = \int_{\mathbb{R}} u'(x)\big[F(x) - G(x)\big]\,dx.$$

Characterizing first-order stochastic dominant cdfs III

Proof (continued).

$$\int_{\mathbb{R}} u(x)\,dG(x) - \int_{\mathbb{R}} u(x)\,dF(x) = \int_{\mathbb{R}} u'(x)\big[F(x) - G(x)\big]\,dx$$

- If $F(x) \geq G(x)$ for all x, the RHS is clearly positive: $G(x) \leq F(x)\,\forall x \implies G \succsim_u F\,\forall u(\cdot)$.
- Suppose there is some x' around which $F(x) < G(x)$; we can then consider a $u(\cdot)$ which is constant except in the neighborhood of x'.
  The RHS will therefore be strictly negative, so there exists a nondecreasing $u(\cdot)$ under which $F \succ_u G$:
  $G(x) \leq F(x)\,\forall x \impliedby G \succsim_u F\,\forall u(\cdot)$. □

Back to our examples. . .

Example

	Lottery				
x	\$95	\$105	\$110	\$80 or \$120	\$90 or \$110
< 80	$F(x) = 0$	0	0	0	0
$[80, 90)$	0	0	0	$1/2$	0
$[90, 95)$	0	0	0	$1/2$	$1/2$
$[95, 105)$	1	0	0	$1/2$	$1/2$
$[105, 110)$	1	1	0	$1/2$	$1/2$
$[110, 120)$	1	1	1	$1/2$	1
≥ 120	1	1	1	1	1

Back to our examples. . .

Example

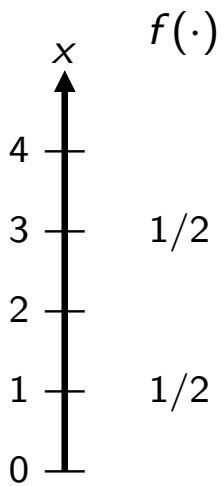
	Lottery				
x	\$95	\$105	\$110	\$80 or \$120	\$90 or \$110
< 80	$F(x) = 0$	0	0	0	0
$[80, 90)$	0	0	0	$1/2$	0
$[90, 95)$	0	0	0	$1/2$	$1/2$
$[95, 105)$	1	0	0	$1/2$	$1/2$
$[105, 110)$	1	1	0	$1/2$	$1/2$
$[110, 120)$	1	1	1	$1/2$	1
≥ 120	1	1	1	1	1

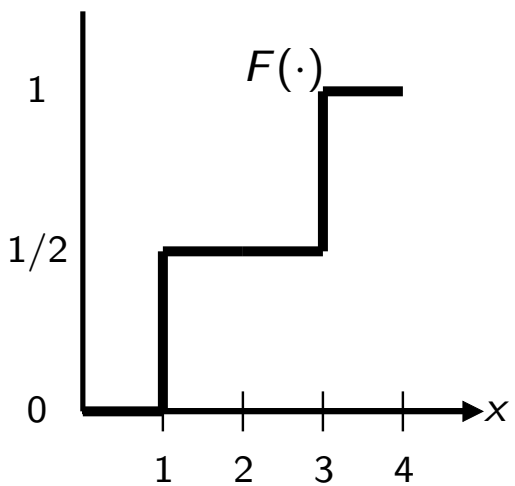
- (\$110) FOSD (\$105) FOSD (\$95)
- (\$110) FOSD (\$90 or \$110)
- Every other combination is ambiguous in terms of FOSD

Characterizing FOSD with upward shifts

Start with a lottery F and construct compound lottery G

- First resolve F
- Then if the resolution of F is some x, hold a second lottery that could potentially increase (but can't decrease) x

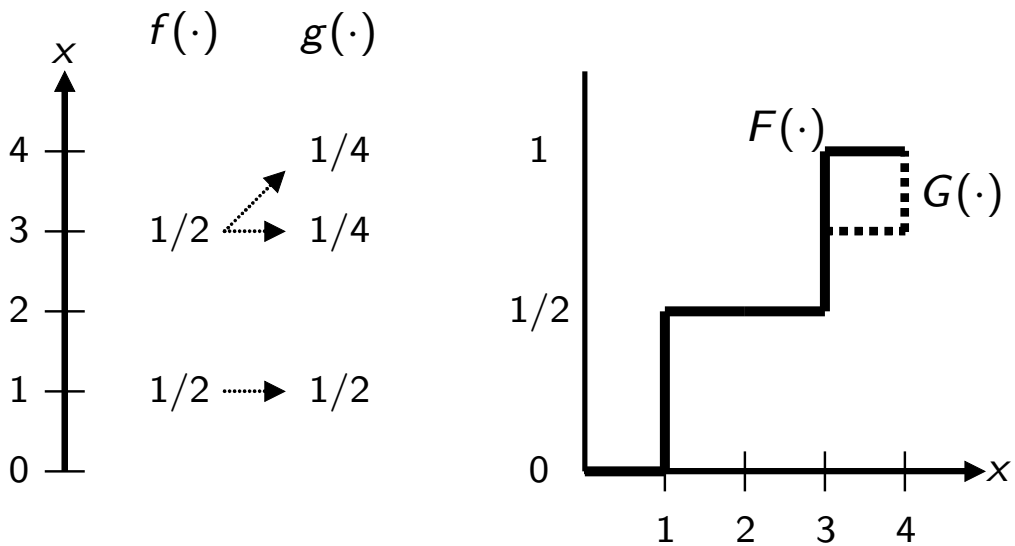




Characterizing FOSD with upward shifts

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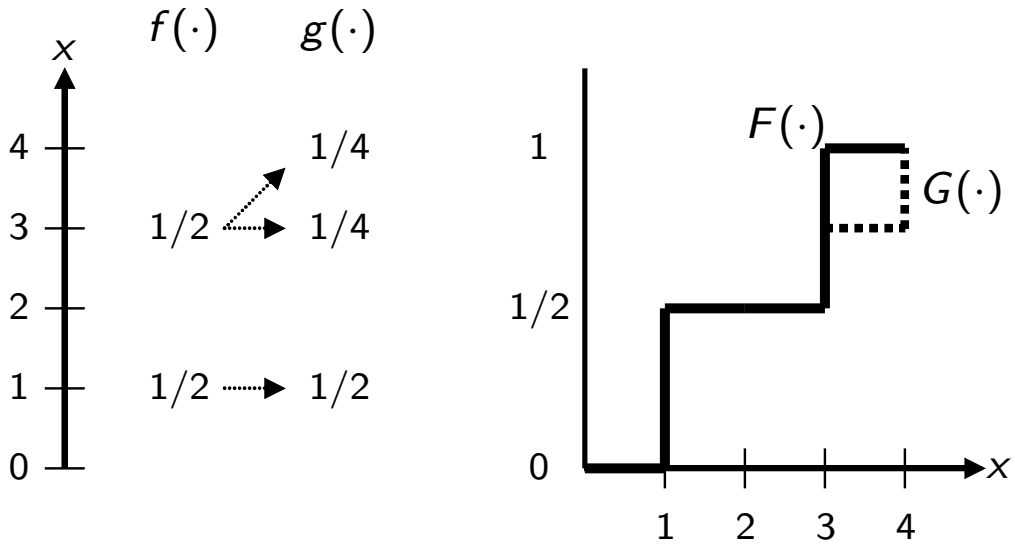
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Characterizing FOSD with upward shifts

Start with a lottery F and construct compound lottery G

- First resolve F
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G FOSD F iff we can construct G from F using upward shifts

Second-order stochastic dominance

FOSD said a lottery was preferred by *all* nondecreasing $u(\cdot)\ldots$

Consider whether a lottery is preferred by all risk-averse $u(\cdot)$

Definition (second-order stochastic dominance)

Suppose F and G have the same mean.

Distribution G second-order stochastic dominates distribution F iff lottery G is preferred to F under every concave, nondecreasing Bernoulli utility function $u(\cdot)$.

That is, for every concave, nondecreasing $u\colon \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}_G\big[u(x)\big] \geq \mathbb{E}_F\big[u(x)\big].$$

Characterizing second-order stochastic dominant cdfs

Theorem

Distribution G second-order stochastic dominates distribution F iff

$$\int_{-\infty}^{x} G(t)\,dt \leq \int_{-\infty}^{x} F(t)\,dt \text{ for all } x.$$

- Proof in notes relies on integration by parts as in FOSD case

Characterizing second-order stochastic dominant cdfs

Theorem

Distribution G second-order stochastic dominates distribution F iff

$$\int_{-\infty}^{x} G(t)\,dt \leq \int_{-\infty}^{x} F(t)\,dt \text{ for all } x.$$

- Proof in notes relies on integration by parts as in FOSD case
- More general proof technique considers basis functions for the class of utility functions
  - Step functions as basis for nondecreasing functions:

$$b_\alpha(x) \equiv \begin{cases} 0, & x \leq \alpha \\ 1, & x > \alpha \end{cases}$$

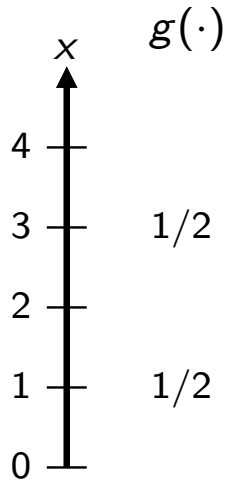
  - Min functions as basis for concave nondecreasing functions:

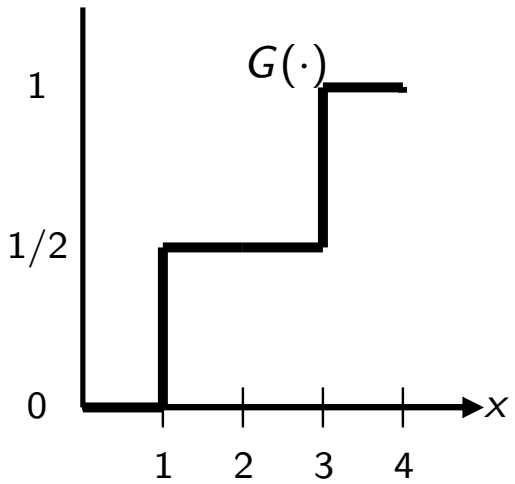
$$b_\alpha(x) \equiv \min\{x, \alpha\}$$

Characterizing SOSD with mean-preserving spreads

Start with a lottery G and construct compound lottery F

- First resolve G
- Then if the resolution of F is some x, hold a second lottery that adds some *zero-mean* random variable to outcome x

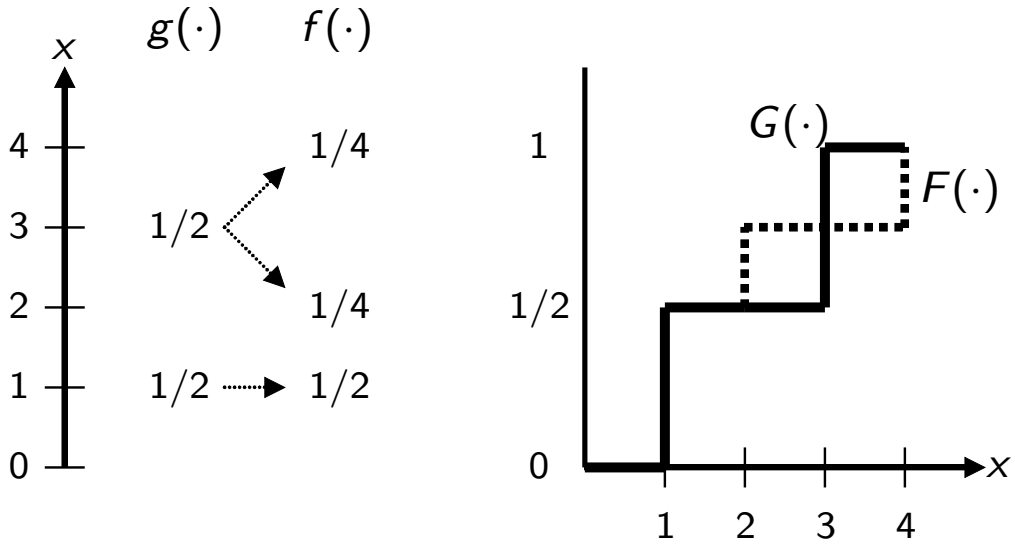




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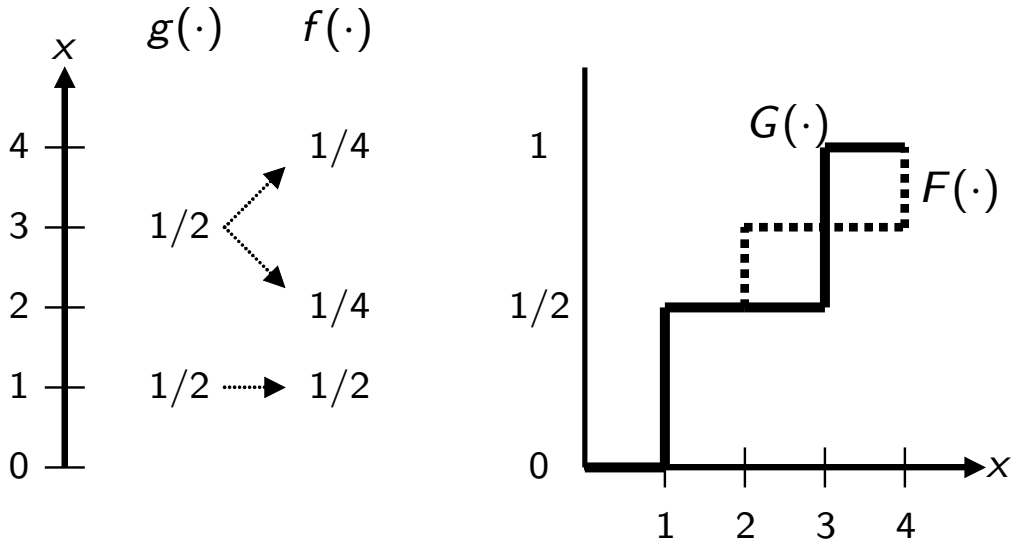
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Characterizing SOSD with mean-preserving spreads

Start with a lottery G and construct compound lottery F

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G SOSD F iff we can construct F from G using mean-preserving spreads

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- Comparing risky prospects
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  - Second-order stochastic dominance
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  - Behavioral criticisms

Demand for insurance: Setup I

- Strictly risk-averse agent with wealth w
- Risk of loss L with probability p
- Insurance available for cost qa pays a in event of loss (agent chooses a)

She solves

$$\max_a pu[w - qa - L + a] + (1 - p)u[w - qa]$$

Demand for insurance: Setup II

The demand for insurance problem

$$\max_a \underbrace{pu[w - qa - L + a] + (1-p)u[w - qa]}_{\equiv U(a)}$$

Note strict concavity of $u(\cdot)$ gives strict concavity of $U(\cdot)$:

$$U'(a) = (1-q)pu'[w - qa - L + a] - q(1-p)u'[w - qa]$$
$$U''(a) = (1-q)^2 pu''[w - qa - L + a] + q^2(1-p)u''[w - qa] < 0$$

Thus FOC is necessary and sufficient:

$$p(1-q)u'[w - qa^* - L + a^*] = q(1-p)u'[w - qa^*]$$

Actuarially fair insurance

What if insurance is actuarially fair?

- That is, insurer makes zero-profit: $q = p$
- FOC becomes

$$\begin{aligned} \cancel{p(1-q)}u'[w - qa^* - L + a^*] &= \cancel{q(1-p)}u'[w - qa^*] \\ w - qa^* - L + a^* &= w - qa^* \\ a^* &= L \end{aligned}$$

- Agent fully insures against risk of loss

Non-actuarially fair insurance

What if insurance is *not* actuarially fair?

- Suppose cost of insurance is above expected loss: $q > p$
- FOC is

$$\begin{aligned}
\frac{u'[w - qa^* - L + a^*]}{u'[w - qa^*]} &= \frac{q(1-p)}{p(1-q)} \\
&> 1 \\
u'[w - qa^* - L + a^*] &> u'[w - qa^*] \\
w - qa^* - L + a^* &< w - qa^* \\
a^* &< L
\end{aligned}$$

- Agent under-insures against risk of loss; it's costly to transfer wealth to the loss state, so she transfers less

A useful comparative statics result I

Consider maximizing a strictly concave differentiable function $U(\cdot)$

FOC: $U'(x^*) = 0$

Note that since $U(\cdot)$ is concave, $U'(\cdot)$ is decreasing

- $U'(x) > 0 = U'(x^*) \iff x < x^*$ (i.e., $U(\cdot)$ is increasing to the left of the maximum)
- $U'(x) < 0 = U'(x^*) \iff x > x^*$ (i.e., $U(\cdot)$ is decreasing to the right of the maximum)

A useful comparative statics result II

Now consider maximizing $U(\cdot, w)$ given a parameter w

FOC: $\partial_x U\big(x^*(w), w\big) = 0$

As above, we know $x < x^*(w)$ iff

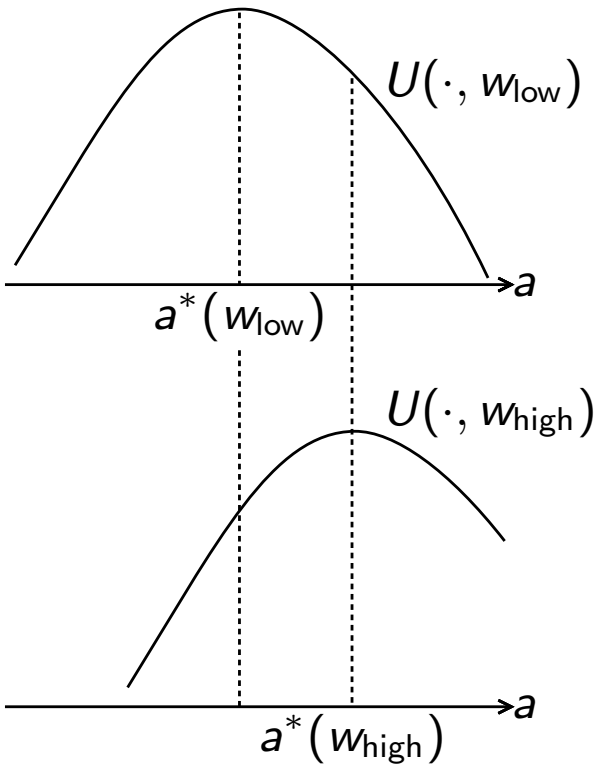
$$\partial_x U(x, w) > 0$$

Letting $x = x^*(w_l)$ for some w_l gives $x^*(w_l) < x^*(w_h)$ iff

$$\partial_x U\big(x^*(w_l), w_h\big) > 0 = \partial_x U\big(x^*(w_l), w_l\big)$$

Suppose the cross partial $\partial_w \partial_x U\big(x^*(w_l), \cdot\big) > 0$; then for all $w_h > w_l$ the above condition holds so $x_*(w_h) > x_*(w_l)$

A useful comparative statics result III



FOCs:

- $\partial_a U\big(a^*(w_{\text{low}}), w_{\text{low}}\big) = 0$
- $\partial_a U\big(a^*(w_{\text{high}}), w_{\text{high}}\big) = 0$

$a^*(w_{\text{low}}) < a^*(w_{\text{high}})$ implies

- $\partial_a U\big(a^*(w_{\text{low}}), w_{\text{high}}\big) > 0$
- $\partial_a U\big(a^*(w_{\text{high}}), w_{\text{low}}\big) < 0$

Sufficient condition is that

$$\partial_w \partial_a U\big(a^*(w_l), \cdot\big) > 0$$

Towards comparative statics in wealth

We showed already that given strict risk-aversion,

- If insurance is actuarially fair ($q = p$), agents fully insure
- If cost of insurance is above expected loss ($q > p$), agents under-insure

Towards comparative statics in wealth

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Intuitively, an agent with decreasing absolute risk aversion will insure less when wealth is higher

Towards comparative statics in wealth

We showed already that given strict risk-aversion,

- If insurance is actuarially fair ($q = p$), agents fully insure
- If cost of insurance is above expected loss ($q > p$), agents under-insure

Intuitively, an agent with decreasing absolute risk aversion will insure less when wealth is higher

$$\begin{aligned} U(a, w) &= pu[w - qa - L + a] + (1 - p)u[w - qa] \\ \partial_w U(a, w) &= pu'[w - qa - L + a] + (1 - p)u'[w - qa] \\ \partial_a \partial_w U(a, w) &= p(1 - q)u''[w - qa - L + a] - q(1 - p)u''[w - qa] \end{aligned}$$

We can't sign this (and use Topkis), but if we can sign it at $a^*(w)$ we can use our previous result...

Towards comparative statics in wealth I

Recall the FOC:

$$p(1-q)u'[\underbrace{w-qa^*-L+a^*}_{\equiv b^*}] = q(1-p)u'[\underbrace{w-qa^*}_{\equiv g^*}]$$

$$\frac{p(1-q)u'[b^*]}{u'[g^*]} = q(1-p)$$

Where b^* is the payout in the "bad" (loss) state, and g^* the payout in the "good" state

By our earlier result, agents under-insure: $g^* > b^*$

Towards comparative statics in wealth II

We can plug in to our earlier expression to get

$$\begin{aligned}
\partial_a \partial_w U(a^*, w) &= p(1-q)u''[b^*] - q(1-p)u''[g^*] \\
&= p(1-q)u'[b^*]\left[\frac{u''[b^*]}{u'[b^*]} - \frac{u''[g^*]}{u'[g^*]}\right] \\
&= p(1-q)u'[b^*]\underbrace{\left[-A[b^*] + A[g^*]\right]}_{<0 \text{ by DARA}} \\
&< 0
\end{aligned}$$

By our "useful comparative statics result,"

When $p < q$, the agent will under-insure; $a^*(w)$ is decreasing in wealth if the agent has DARA and increasing in wealth if the agent has IARA.

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The portfolio problem: Setup

- Risk-averse agent with wealth w
- Choice of investment across two assets
  1. Safe asset returns r for certain
  2. Risky asset returns z distributed according to cdf $F(\cdot)$, with a higher expected return than the safe asset; i.e., $\mathbb{E}_F z > r$

The portfolio problem: Setup

- Risk-averse agent with wealth w
- Choice of investment across two assets
  1. Safe asset returns r for certain (investment $w - a$)
  2. Risky asset returns z distributed according to cdf $F(\cdot)$, with a higher expected return than the safe asset; i.e., $\mathbb{E}_F z > r$ (investment a)

Agent winds up with $az + (w - a)r$

She solves

$$\max_a \underbrace{\mathbb{E}_F\left[u(az + (w-a)r)\right]}_{U(a)} \equiv \max_a \underbrace{\int_{\mathbb{R}} u\big(az + (w-a)r\big)\, dF(z)}_{U(a)}$$

The portfolio problem: Setup

- Risk-averse agent with wealth w
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  1. Safe asset returns r for certain (investment $w - a$)
  2. Risky asset returns z distributed according to cdf $F(\cdot)$, with a higher expected return than the safe asset; i.e., $\mathbb{E}_F z > r$ (investment a)

Agent winds up with $az + (w - a)r$
She solves

$$\max_a \underbrace{\mathbb{E}_F\left[u(az + (w-a)r)\right]}_{U(a)} \equiv \max_a \underbrace{\int_{\mathbb{R}} u\big(az + (w-a)r\big)\, dF(z)}_{U(a)}$$

First-order condition:

$$\int_{\mathbb{R}} (z - r) u'\big(a^* z + (w - a^*)r\big)\, dF(z) = 0$$

Risk-neutral agent

If agent is risk-neutral, $u(x) = \alpha x + \beta$; the FOC would require

$$\int_{\mathbb{R}} (z - r) u'\left(a^* z + (w - a^*) r\right) dF(z) \stackrel{?}{=} 0$$

$$\int_{\mathbb{R}} (z - r) \alpha \, dF(z) \stackrel{?}{=}$$

$$\int_{\mathbb{R}} (z - r) \, dF(z) \stackrel{?}{=}$$

$$\mathbb{E}_F z \neq r$$

We have a corner solution: agent puts all investment in risky asset
A risk-neutral investor only cares about *expected* return

Strictly risk-averse agent: intuition from the FOC

If agent is strictly risk-averse, strict concavity of $u(\cdot)$ gives strict concavity of $U(\cdot)$: FOC is necessary and sufficient

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Could $a^* = 0$ be a solution?

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Could $a^* = 0$ be a solution? Only if

$$\int_{\mathbb{R}} (z - r) u'(wr) \, dF(z) \stackrel{?}{=} 0$$

$$\int_{\mathbb{R}} (z - r) \, dF(z) \stackrel{?}{=} 0$$

$$\mathbb{E}_F z \neq r$$

A strictly risk-averse investor will always invest at least a little in the risky asset!

A useful comparative statics result

Recall our earlier result: when maximizing a strictly concave differentiable function $U(\cdot)$,

- $U'(a) > 0 = U'(a^*) \iff a < a^*$ (i.e., $U(\cdot)$ is increasing to the left of the maximum)
- $U'(a) < 0 = U'(a^*) \iff a > a^*$ (i.e., $U(\cdot)$ is decreasing to the right of the maximum)

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Therefore if $U'(a) = 0 \implies V'(a) \geq 0$, we know that

$$\operatorname*{argmax}_a U(a) \leq \operatorname*{argmax}_a V(a).$$

Comparative statics in risk aversion I

Suppose agent v is less risk-averse than agent u

- We expect v to invest more in the risky asset
- It is a sufficient condition for $U'(a) = 0 \implies V'(a) \geq 0$

We can write $v(x) = h\big(u(x)\big)$ for some nondecreasing *convex* $h(\cdot)$

$$U(a) \equiv \int_{\mathbb{R}} u\big(az + (w-a)r\big)\, dF(z)$$

$$V(a) \equiv \int_{\mathbb{R}} v\big(az + (w-a)r\big)\, dF(z)$$

$$= \int_{\mathbb{R}} h\big[u\big(az + (w-a)r\big)\big]\, dF(z)$$

$$V'(a) = \int_{\mathbb{R}} h'\big[u\big(az + (w-a)r\big)\big]\, u'\big(az + (w-a)r\big)(z-r)\, dF(z)$$

Comparative statics in risk aversion II

Consider evaluating $V'(a^*)$ where a^* satisfies $U'(a^*) = 0$
Consider the term $h'\left[u\left(a^*z + (w - a^*)r\right)\right]$, which is increasing in z since $h(\cdot)$ is convex

- Define $\tilde{h}$ as this term evaluated when $z = r$; note $\tilde{h} \geq 0$
- When $z \leq r$, the term is below $\tilde{h}$
- When $z \geq r$, the term is above $\tilde{h}$

Comparative statics in risk aversion III

$$V'(a^*) = \int_{\mathbb{R}} h'\big[u\big(a^*z+(w-a^*)r\big)\big]\,u'\big(a^*z+(w-a^*)r\big)(z-r)\,dF(z)$$

$$= \int_{-\infty}^{r} \underbrace{h'\big[u\big(a^*z+(w-a^*)r\big)\big]}_{\leq \tilde{h}}\,u'\big(a^*z+(w-a^*)r\big)\,\underbrace{(z-r)}_{\leq 0}\,dF(z)$$

$$+ \int_{r}^{\infty} \underbrace{h'\big[u\big(a^*z+(w-a^*)r\big)\big]}_{\geq \tilde{h}}\,u'\big(a^*z+(w-a^*)r\big)\,\underbrace{(z-r)}_{\geq 0}\,dF(z)$$

$$\geq \int_{-\infty}^{r} \tilde{h}u'\big(a^*z+(w-a^*)r\big)(z-r)\,dF(z)$$

$$+ \int_{r}^{\infty} \tilde{h}u'\big(a^*z+(w-a^*)r\big)(z-r)\,dF(z)$$

$$= \int_{\mathbb{R}} \tilde{h}u'\big(a^*z+(w-a^*)r\big)(z-r)\,dF(z) = \tilde{h}U'(a^*) = 0$$

Comparative statics in risk aversion: what have we done?

- We showed that $U'(a) = 0 \implies V'(a) \geq 0$
- Therefore $\text{argmax}_a\, U(a) \leq \text{argmax}_a\, V(a)$
- This confirms our "intuition" that a less risk-averse agent invests more in the risky asset

The notes use the same logic to argue that an agent with DARA will invest more in the risky asset at higher levels of wealth

Similar results can be proven for DRRA

Outline

- Comparing risky prospects
  - First-order stochastic dominance
  - Second-order stochastic dominance
- Application: demand for insurance
- Application: the portfolio problem
- Beyond the expected utility model
  - Subjective probability
  - Behavioral criticisms

Subjective probabilities

We have assumed the decision-maker accurately understands the likelihood of each outcome

- Set of outcomes $\mathcal{X}$ (e.g., dollar payouts)

Subjective probabilities

We have assumed the decision-maker accurately understands the likelihood of each outcome; suppose not

- Set of outcomes $\mathcal{X}$ (e.g., dollar payouts)
- Set of "states of the world" $\mathcal{S}$
- Bets (a.k.a. "acts") are mappings from states of the world to outcomes $f : \mathcal{S} \to \mathcal{X}$

Savage shows that if preferences over acts satisfy certain axioms, they must be *as if* maximizing a vN-M utility function given

- Some probability distribution over $\mathcal{S}$
- A Bernoulli utility function $u : \mathcal{X} \to \mathbb{R}$

Problems with the independence axiom

Example

Which lottery would you rather face?

	\$0	\$48	\$55
Lottery A	1%	66%	33%
Lottery B		100%	

Problems with the independence axiom

Example

Which lottery would you rather face?

	\$0	\$48	\$55	Expected payout
Lottery A	1%	66%	33%	\$50
Lottery B		100%		\$48

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Example

Which lottery would you rather face?

	\$0	\$48	\$55	Expected payout
Lottery C	67%		33%	\$18
Lottery D	66%	34%		\$16

Problems with the independence axiom

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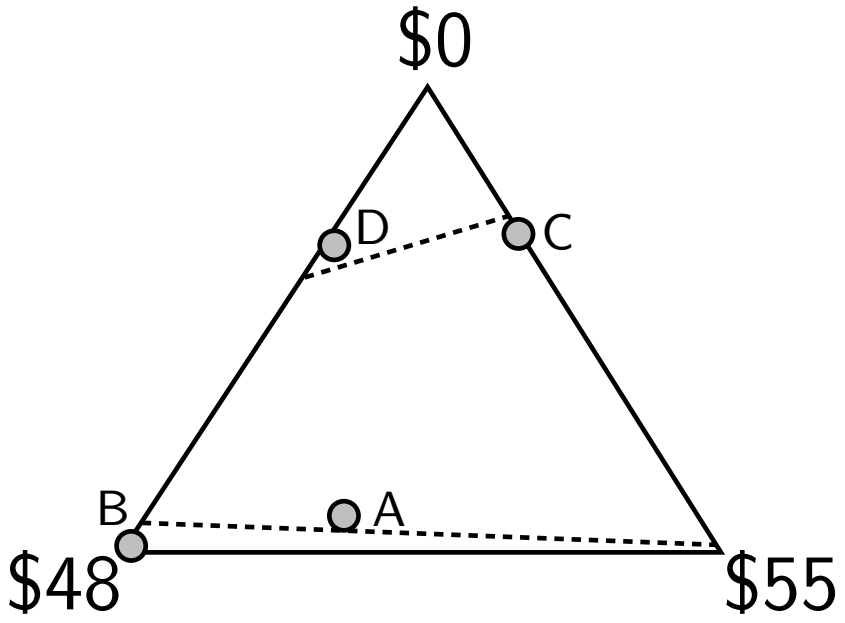
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Illustrating Allais' experiment

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Problems with risk aversion

It seems like people are way too risk averse on small-stakes gambles

Example

Would you bet on a fair coin toss where you lose \$1000 or win \$1050?

Problems with risk aversion

It seems like people are way too risk averse on small-stakes gambles

Example

Would you bet on a fair coin toss where you lose \$1000 or win \$1050?

If you would always turn down such a bet (at any wealth level), you would turn down a bet on a fair coin where you lose \$20,000 or gain *any* amount

"Loss aversion" has been suggested as an explanation

Framing experiment 1

Example

The U.S. is preparing for an outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Scientists predict that:

- If program A is adopted, 200 people will be saved.
- If program B is adopted, there is a $^2/_3$ chance that no one will be saved, and a $^1/_3$ probability that 600 people will be saved.

Which program would you choose?

Framing experiment 1

Example

The U.S. is preparing for an outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Scientists predict that:

- If program A is adopted, 200 people will be saved. [72%]
- If program B is adopted, there is a $^2/_3$ chance that no one will be saved, and a $^1/_3$ probability that 600 people will be saved. [28%]

Which program would you choose?

Framing experiment 2

Example

The U.S. is preparing for an outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Scientists predict that:

- If program C is adopted, 400 people will die with certainty.
- If program D is adopted, there is a $^2/_3$ probability that 600 people will die, and a $^1/_3$ probability that no one will die.

Which program would you choose?

Framing experiment 2

Example

The U.S. is preparing for an outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Scientists predict that:

- If program C is adopted, 400 people will die with certainty. [22%]
- If program D is adopted, there is a $^2/_3$ probability that 600 people will die, and a $^1/_3$ probability that no one will die. [78%]

Which program would you choose?

These two examples are exactly the same question stated in different ways!